Physica A 176 (1991) 117–137
North-Holland

ORDERED PHASES OF DNA IN VIVO AND IN VITRO

Françoise LIVOLANT
Centre de Biologie Cellulaire (CNRS), 67 rue Maurice Günsbourg, 94200 Ivry-sur-Seine, France

In vitro, pure DNA forms multiple liquid crystalline phases when the polymer concentration is increased: precholesteric organization, cholesteric phase and columnar hexagonal phase. Similar organizations of chromatin can be found in vivo: hexagonal packing in bacteriophages and certain sperm heads, cholesteric organization in dinoflagellate chromosomes, bacterial nucleoids and mitochondrial DNA, helical-shaped chromosomes in many species. The different forms of condensed chromatin seem to be related to different local concentrations of DNA. In the highly condensed forms, chromatin is inactive and the double stranded DNA molecule is linear with small amounts of associated proteins.

A more detailed analysis is presented in the case of cholesteric structures (helical pitch, defects) in polarizing microscopy and in electron microscopy. Differences observed in vitro and in vivo are probably related to the length of the DNA molecule and to the presence of proteins associated to DNA in the chromosomes.

1. Introduction

The DNA molecule is densely packed in vivo in chromosomes, sperm nuclei and virus capsids and the local DNA concentration can reach values up to 800 mg/ml [1]. Pure DNA forms in vitro highly ordered structures in the same range of concentration, namely liquid crystalline phases. We present here a comparative analysis of these condensed forms of DNA in vivo and in vitro.

2. Condensed forms of chromatin

2.1. Cholesteric structure

Many biological materials present in sections series of nested arcs that reveal a fibrillar cholesteric organization, as interpreted by Bouligand et al. [2] and explained in fig. 1. These series of arcs are drawn by the projection onto the section plane of the successive orientations of molecules along the cholesteric axis. They can be seen when the section plane is oblique with respect to the cholesteric stratification and one series of arcs corresponds to a rotation of 180°

0378-4371/91/$03.50 © 1991 – Elsevier Science Publishers B.V. (North-Holland)

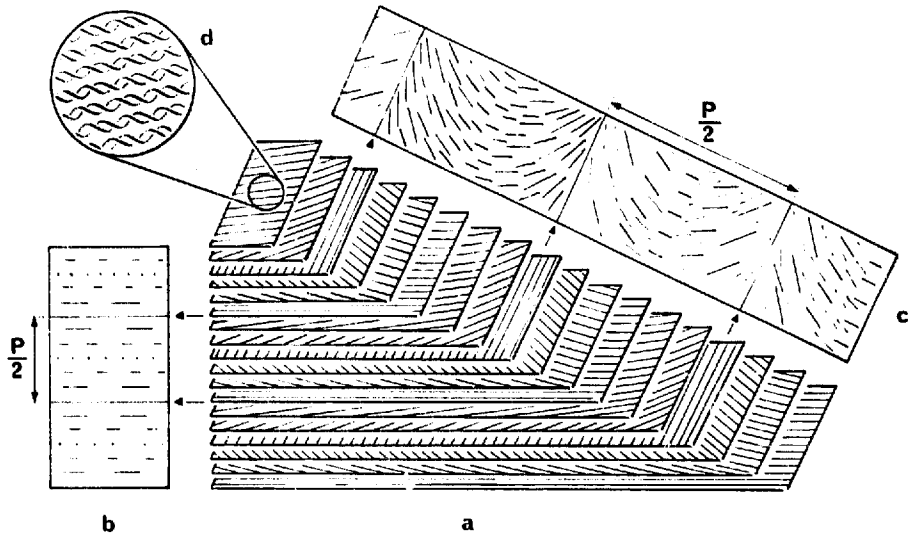


Fig. 1. Model of the cholesteric organization (a) with the different patterns observed in sections (b, c, d). The DNA molecules (enlarged in d) are aligned in parallel and their orientation rotates from plane to plane. Section planes parallel to the cholesteric axis (b) reveal periodic patterns with the alternation of filaments parallel, oblique and normal to the section plane (represented by lines, segments and points). The periodicity corresponds to a rotation of 180° of the molecular orienations ($P/2$). In oblique sections (c), the different orientations of the filaments create series of arcs and the periodicity corresponds to $P/2$. In sections normal to the cholesteric axis (d) filaments are aligned in parallel.

of the molecular orientations, i.e. to half the helical pitch $P/2$ (fig. 1c). In sections nearly parallel to the cholesteric stratification (fig. 1d), filaments seem to be aligned more or less in parallel while sections parallel to the cholesteric axis present a periodicity without arced patterns, with an alternation of filaments parallel, oblique and perpendicular to the observation plane (fig. 1b). These periodic patterns devoid of arcs make it possible to measure accurately the half-helical pitch $P/2$ in the structure.

A well known example is given by dinoflagellates, which are primitive unicellular algae. In the nucleus, the main part of chromatin remains condensed throughout the cell cycle in the form of elongated chromosomes. The different patterns described in fig. 1 can be recognized: parallel alignment of chromatin filaments and periodic patterns with and without series of arcs (fig. 2a). They correspond to different orientations of the chromosome axis (and cholesteric axis) relative to the section plane. Chromosomes are surrounded by the nucleoplasm in which loops of extrachromosomal DNA can be followed. The relative amount of DNA in chromosomes and nucleoplasm seems to vary during the cell cycle but there are no precise data available about the dynamics of this phenomenon.

The arced patterns are also observed in various species of bacteria (*Escherichia coli*, *Bacillus subtilis*, *Rhizobium*, etc. [3–9]), but they are rarely

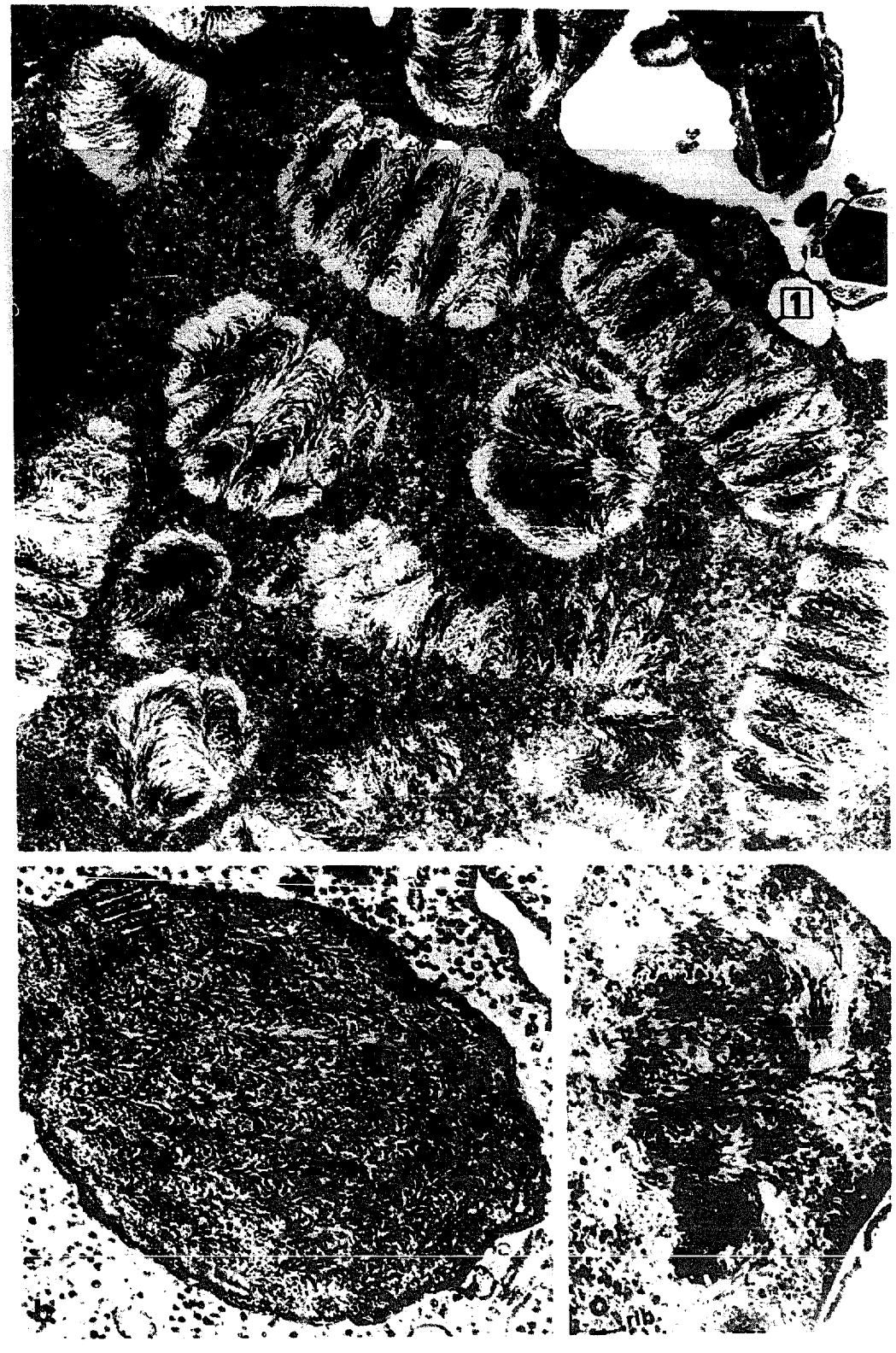


Fig. 2. Typical arced patterns in cholesteric chromatin (thin sections observed in electron microscopy).

(a) Dinoflagellate chromosomes. The different patterns (1, 2, 3) correspond to different orientations of the section plane with respect to the cholesteric axis, as explained in fig. 1 (×30 000).

(b) Mitochondrial DNA (k DNA) in a ciliate (*Bodo*) (×48 000). Reprinted from Brugerolle and Mignot (1977) [15].

(c) Bacterial nucleoïd in *Rhizobium* (×60 000). Reprinted from Gourret (1978) [8].

interpreted as biological analogues of cholesteric liquid crystals. Besides, this organization is not always present. In several species, nucleoids are decondensed during the exponential growth stage, in which an active synthesis of DNA occurs, while they are condensed and show a cholesteric organization during the inactive stationary phase. In *Rhizobium*, two types of cells are observed: rapid-growth bacteria with dispersed nucleoids and slow-growth bacteria (in which protein synthesis and consequently transcription is reduced) with condensed nucleoids showing series of nested arcs (fig. 2c) [8]. Other bacteria do not show this organization in normal culture conditions but this organization can be induced by addition of inhibitors of the protein synthesis in the culture medium [10–14].

A cholesteric organization of mitochondrial DNA has also been described in certain flagellate protozoa. In these cells, mitochondria contain a large amount of circular DNA molecules (k DNA) arranged in a structure which is called the kinetoplast. In *Bodo*, this DNA is usually decondensed but, when the cells become encysted, their k DNA condenses progressively and series of nested arcs become visible in sections (fig. 2b) [15].

A cholesteric organization is also found in sperm nuclei from many species. During the maturation of the sperm cell, many changes occur in the chemical composition of chromatin, namely the nature and relative amounts of proteins associated to DNA, and the activity of chromatin reduces progressively. In the same time, chromatin condenses progressively. A cholesteric organization of chromatin can be found in the last stages of this evolution (spermatids and mature sperm nuclei) but, contrary to the first examples, the arced patterns are rarely observed. In most mammal sperm nuclei (rat, rabbit, bull, stallion), a lamellar structure with a periodicity ranging from 250 to 330 Å is observed on freeze–fracture replicas [16–20]. Besides, chromatin extracted from stallion sperm nuclei shows circular dichroism properties characteristic of a cholesteric organization [21]. The periodicity revealed by freeze–fracture in the intact sperm nuclei corresponds probably to the cholesteric stratification but the half-helical pitch is too small (250–300 Å) to make the series of nested arcs visible in sections about 500 Å thick. Chromatin is probably arranged in a planar cholesteric structure with the cholesteric axis perpendicular to the flattening plane of the sperm head. The birefringence properties of the sperm nuclei strongly supports this hypothesis [20]. A periodic stratification of about 350 Å is also seen in spermatids from pleurodele [22] and series of nested arcs were occasionally observed (Picheral, unpublished results). In this case, the shape of the nucleus is highly elongated and the cholesteric structure of chromatin is cylindrical. Other examples of twisted chromatin can be found in sperm cells. In the whip scorpion [23], thick filaments of chromatin are aligned

in parallel and a twist appears between them. It is a double-twist cholesteric structure and the half-helical pitch would be larger than the diameter of the nucleus (about 1 μm). As the condensation process is going on, a progressive coiling of the nucleus can be followed. This coiling corresponds probably to a relaxation of the frustration induced by the double-twist cholesteric structure as already described in other biological structures [24]. The same type of chromatin organization is also found in a fish (*Scyliorhinus caniculus*) [25]. In other sperm nuclei, we do not have any information on the chromatin structure but a twist can be suspected because the nuclei present very beautiful helical shapes, for example in the green tree frog (*Rhacophorus*) [26] and in the octopus (*Eledone cirrhosa*) [27].

2.2. Hexagonal packing of DNA

A hexagonal packing of DNA molecules has been described in other sperm nuclei (sepia, trout and salmon) [28, 29] and in numerous bacteriophages (*T2, T4, T7, T22, Lambda, φ29*) [30–33]. In these bacteriophages, the DNA interhelix spacing obtained from X-ray diffraction analyses varies from 26.4 Å to 28 Å. This regular packing was assessed in *T4* bacteriophages by cryoelectron microscopy of vitrified samples [34].

During the encystment of the ciliate *Bursaria*, part of the inactive chromatin condenses into crystal-like structures in the form of hexagonal platelets [35, 36]. We lack precise information about the organization of chromatin filaments inside these crystals.

2.3. Smectic-like structure

In another flagellate (*Trypanosoma*), the kinetoplast (previously defined) appears as a curved disc structure, 1–1.5 μm in diameter and 0.16 μm thick. It is made of about 24 000 circular DNA molecules 0.45 μm long. Its aspect is that of a bilayer and it would be formed by the parallel alignment of supertwisted DNA minicircles [37].

The addition in the growing medium of ethidium bromide, which is known to intercalate between the base pairs in the DNA molecule, to modify the helical pitch of the double helix and its superhelicity, changes the structure of the kinetoplast. The bilayer structure disappears and series of nested arcs become visible. The smectic-like organization turns to cholesteric [38]. It is probably the change in the superhelicity of DNA which induces the supramolecular ordering transition.

2.4. Helical-shaped chromosomes and nuclei

It is often difficult to get information about the organization of chromatin filaments in highly condensed structures. Nevertheless one encounters very frequently helical shaped nuclei or chromosomes and we can guess that the global helical shape reveals a local order between DNA molecules. We already described the helical shape of certain sperm nuclei, from the green tree frog [26], the octopus [27], certain fish [25] and the whip scorpion [23] and we suggested at least in some of them the existence of a local twist between DNA molecules. Many chromosomes from eucaryotes present also a helical shape: mitotic human chromosomes [39–41], chromosomes from triton and carp erythrocyte nuclei and from rabbit leucocyte nuclei [42] and meiotic chromosomes from a plant (*Tradescantia*) [43]. Their helical structure is often emphasized either by a pretreatment with acetic acid, which extracts partially some components of chromatin, or revealed by a selective digestion of RNA [44].

2.5. Characteristics of condensed chromatin

To summarize, the different examples of highly ordered condensed chromatin are gathered in table I and we distinguished linear and circular DNA molecules.

The different packing geometries seem to correspond to different degrees of compaction of chromatin, first helical, then cholesteric (with half-helical pitches going from more than 2000 Å to 250 Å), and finally hexagonal. The smectic-like order would be an answer to particular conditions of circularity and supertwist of the DNA molecule.

2.5.1. Structure of chromatin filament

The helical eucaryotic chromosomes, which represent here the "less concentrated" chromatin, must be considered separately. First, we do not have precise information about the local organization of the chromatin filament. Second, their chromatin is different from that of other condensed DNA structures presented here. It is a typical eucaryotic chromatin in which basic proteins, called histones, are associated to DNA to form the nucleosomic structure. When decondensed and spread, the chromatin filaments appear as "beads on a string". Each bead, or nucleosome, is made of a 146 base pairs length of DNA coiled around a core of histones. A spacer DNA (40 base pairs long) join two nucleosomes.

In all other condensed forms of chromatin, the linearity of the DNA molecule is kept, whatever the nature and amount of proteins associated with DNA (but there are never classical histones). In chromatin from spermatids

Table I
Condensed forms of chromatin in vivo.

	Long DNA molecules	Circular DNA molecules
Helical structures	sperm nuclei (*Octopus*, *Rhacophorus*, *Eledone*) chromosomes[a] (*Tradescantia*, *human*, *rabbit*, *carp*)	
Cholesteric	dinoflagellate chromosomes bacterial nucleoids sperm nuclei (*stallion*, *rat*, *rabbit*, *bull*, *pleurodele*, *whip scorpion*)	mitochondrial DNA (*Bodo*)
"Smectic"		Mitochondrial DNA (*Trypanosoma*)
Hexagonal packing	bacteriophages (*T2*, *T4*, *T7*, *Lambda*, *ϕ29*) sperm nuclei (*sepia*, *trout*, *salmon*) ciliate chromatin (*Bursaria*)	

[a] Chromatin with a nucleosomic structure.

and mature sperm cells, histones have been replaced by other proteins which are located in the small grove of the DNA molecule [45]. Chromatin from dinoflagellates, bacteria and kinetoplasts is devoid of classical histones. The chromatin filament is smooth.

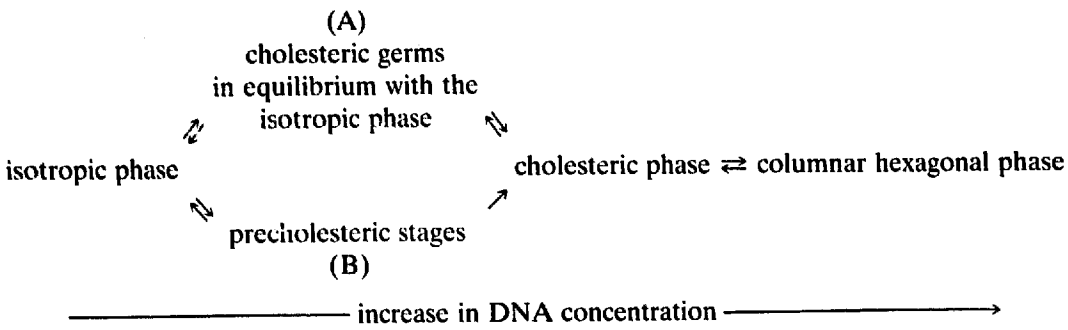
2.5.2. *Inactivity of DNA*

In condensed forms of chromatin, the DNA molecule is inactive. There is no duplication and no transcription (copy of the genetic information into proteins). In dinoflagellates, the major part of the DNA is inactive and located in the cholesteric chromosomes. The small portion of active DNA is decondensed and forms loops extruding from the chromosomes into the nucleoplasm. Transcription activity has been localized along these loops by autoradiography [46]. We saw that the cholesteric forms of bacterial DNA are found in cells where the protein synthesis is low or inhibited experimentally. In sperm nuclei, chromatin is classically known to be inactive.

The dense packing of DNA seems to be a solution to store the genetic material into a small volume and to preserve its integrity at a moment of the cell cycle or of the life of the organism. The packing solution must maintain the ability for DNA to decondense and to enter a new phase of decondensation and activity.

3. Liquid crystalline phases of DNA

In aqueous solution, DNA forms multiple liquid crystalline phases [47–57] and their nature depends on the polymer concentration. The evolution can be summarized as follows:



The transition from the dilute and isotropic phase to the cholesteric structure can follow two different evolutions. In the first one (A), there is formation of small cholesteric germs in the isotropic phase, which grow and merge to form finally an homogeneous cholesteric phase. The second one (B) corresponds to a higher order transition and seems to affect exclusively long molecules [51]. This transition follows a succession of stages that we call precholesteric. For highest concentrations, the cholesteric phase transforms into a columnar hexagonal phase [57].

Up to now, the structure of these phases was essentially analyzed in polarizing microscopy. Their ultrastructure is now investigated with the resolution of electron microscopy by using freeze–fracturing techniques. A short summary of the structural data available will be given here.

3.1. Cholesteric phase

When observed in polarizing microscopy, the cholesteric phase presents usually typical fingerprint patterns (fig. 3a). The periodicity, with alternation of light and dark stripes, corresponds to half the helical pitch, as explained in fig. 3c.

Freeze–fracture electron microscopy of this phase reveals three different typical patterns, which were shown to correspond to three different orientations of the fracture plane relative to the cholesteric axis [58]. The most frequent one shows series of nested arcs (fig. 3b) and these series of arcs can be compared to those obtained in the section shown in fig. 1c. They correspond to a rotation of 180° of the molecular orientations ($P/2$). It is therefore possible to correlate the electron microscopic patterns to the fingerprint patterns. Textures

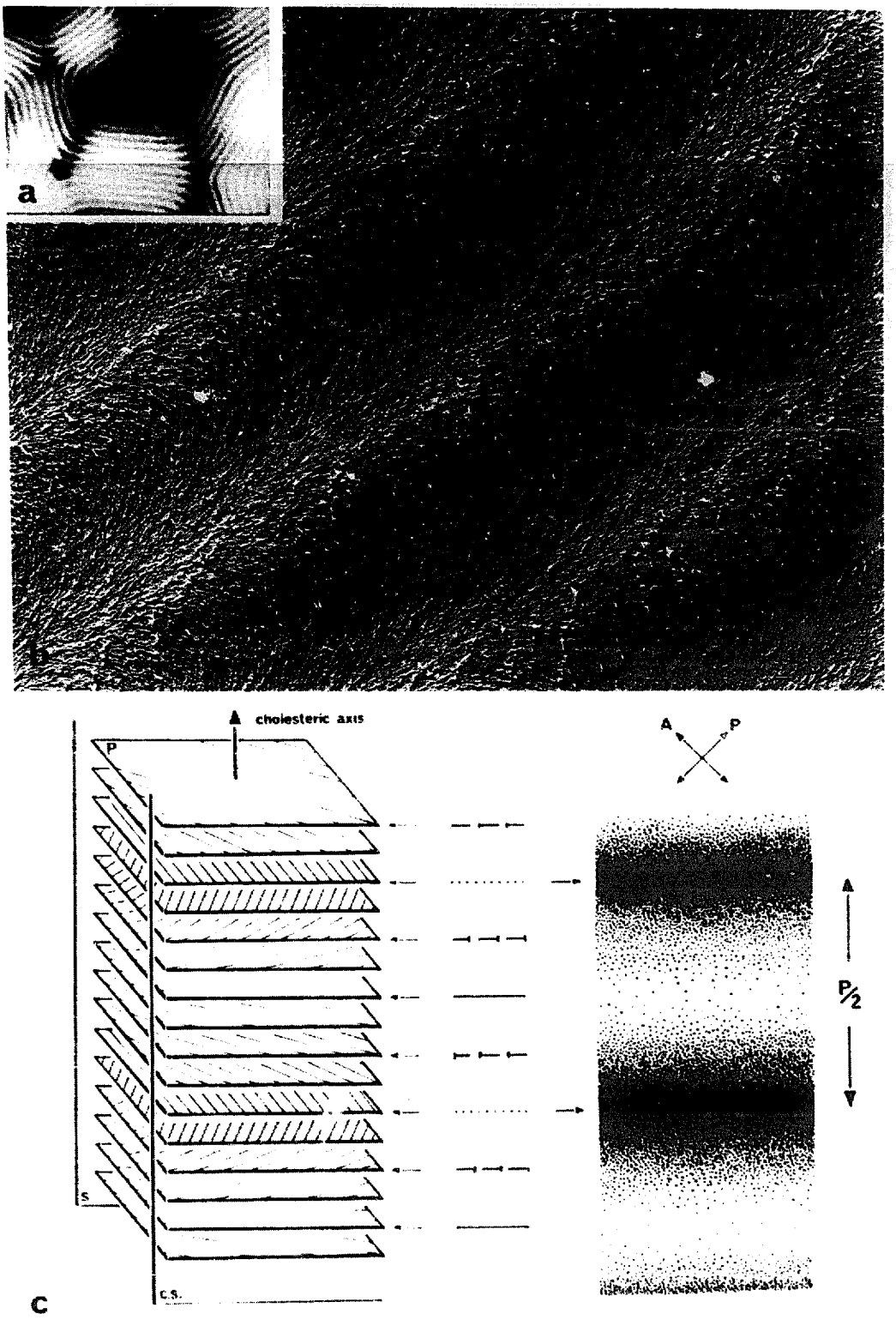


Fig. 3. Cholesteric liquid crystalline phase of DNA.
(a) Typical fingerprint patterns observed in polarizing microscopy (crossed circular polars, ×1000).
(b) Series of nested arcs visible in electron microscopy after freeze–fracture (×11 700, courtesy of A. Leforestier).
(c) Schematic representation of the cholesteric organization. When the cholesteric axis lies parallel to the plane of slide (s) and cover slip (c.s.), molecules are alternatively parallel, oblique and normal to the observation plane, which is represented by lines, nails and points. Between crossed polars (P, A), the light is transmitted when molecules lie parallel to the observation plane and the cholesteric structure appears as an alternation of light and dark bands. The periodicity corresponds to half the helical pitch ($P/2$).

and defects of the cholesteric phase can be analyzed in optical and electron microscopy (work in progress, in collaboration with A. Leforestier).

3.2. Cholesteric germs

Cholesteric germs of DNA can be divided into two groups. There are spherical or elongated germs, which show a uniform orientation of the cholesteric layers (fig. 7a), and germs with concentric layers, which are always spherical. The shape of the germ usually reflects the behaviour of molecules at the interface between the cholesteric and the isotropic phase. Molecules exclusively parallel to the interface lead to the formation of spherical germs whereas different possible orientations at the interface favour the elongation of the germs. More complex germs can be formed by the fusion of several elementary germs [52]. The smallest germs growing in the isotropic phase cannot be seen in polarizing microscopy, but they can be detected in electron microscopy. Moreover, the molecular behaviour at the interface can be investigated more precisely with this method (work in progress, in collaboration with A. Leforestier).

3.3. Precholesteric organization of DNA

Long DNA molecules (0.05–1.9 μm) obtained by sonication can follow another transition from the isotropic to the cholesteric phase, which we called precholesteric [51]. There is no clear interface between the two phases but a continuous transition that we divided into five steps. Molecules seem to align to form a long range helical structure successively in the three directions of space. A typical step of this evolution showing clearly its helical nature is presented in fig. 4a. Finally, the cholesteric organization appears locally in many places and grows from this helical structure. We think that this organization is specific of long molecules since it was never observed with short DNA fragments (500 Å in length).

3.4. Columnar hexagonal phase

The more concentrated phase was shown to be columnar hexagonal when made of DNA fragments about 500 Å in length (Livolant et al., 1989) or of longer DNA fragments heterogeneous in length (0.05–1.9 μm) [53]. We cannot rule out the occurrence of a smectic phase in case of short DNA fragments extremely well defined in length, but up to now there is no evidence of such a phase. The columnar nature of this phase was demonstrated by the use of three different methods: analysis of the textures in polarizing microscopy, freeze–fracture electron microscopy and X-ray diffraction [57].

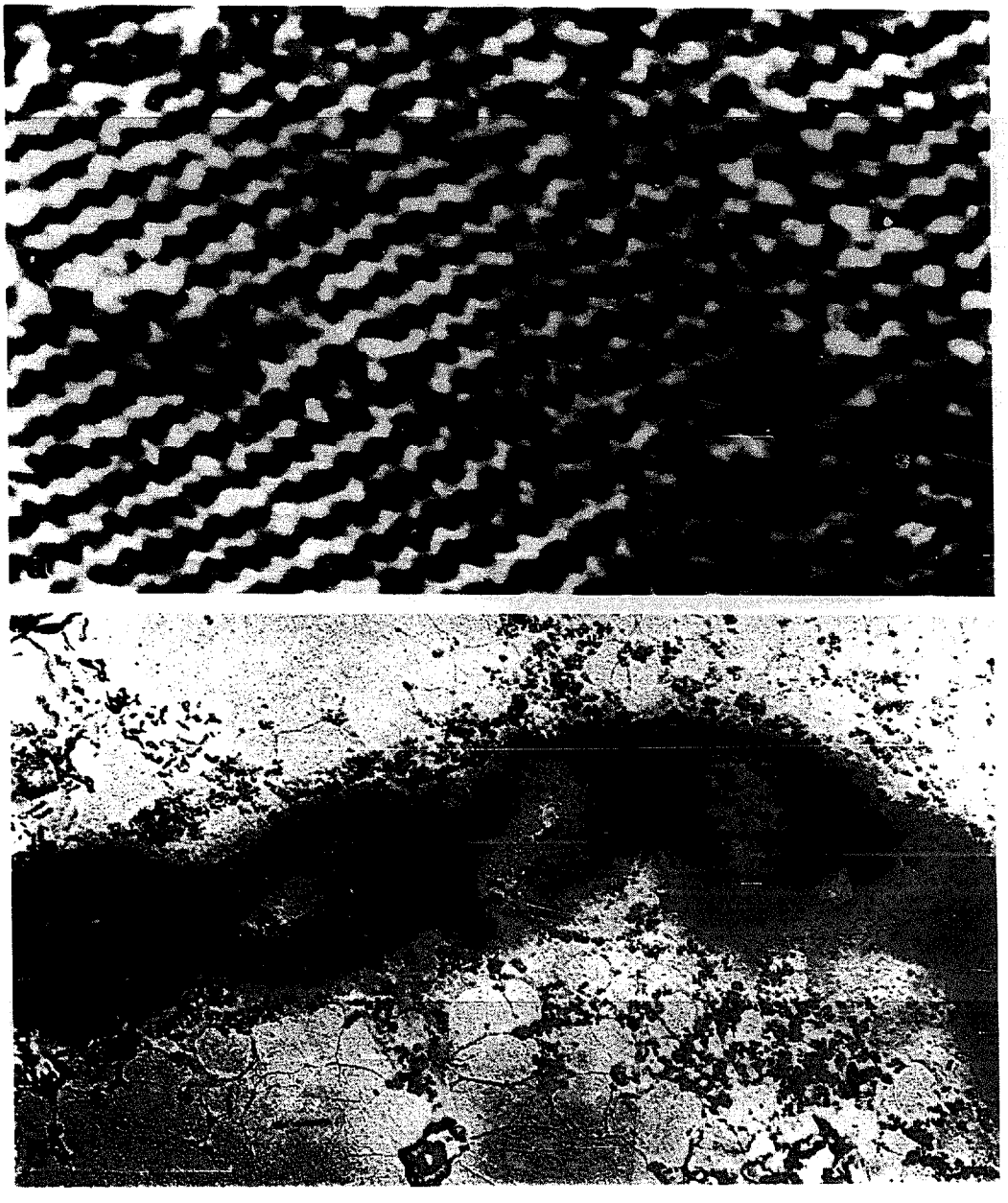


Fig. 4. Helical organization of DNA in vitro and in vivo.
(a) Precholesteric organization of long DNA molecules (crossed polars, ×1800).
(b) Spread dinoflagellate chromosome (observation in electron microscopy, ×19 000).

Freeze–fracture experiments reveal that this phase behaves as a lamellar structure (fig. 5b). Each layer (about 28–40 Å thick) is made of molecules aligned in parallel and the molecular orientation remains unchanged in the successive superimposed layers. The molecular orientation in each layer can be easily determined by looking at the furrows left by some molecules which have been teared out during the fracturing process. Double stranded DNA molecules themselves can be seen individually at the limit of each fragment of layer

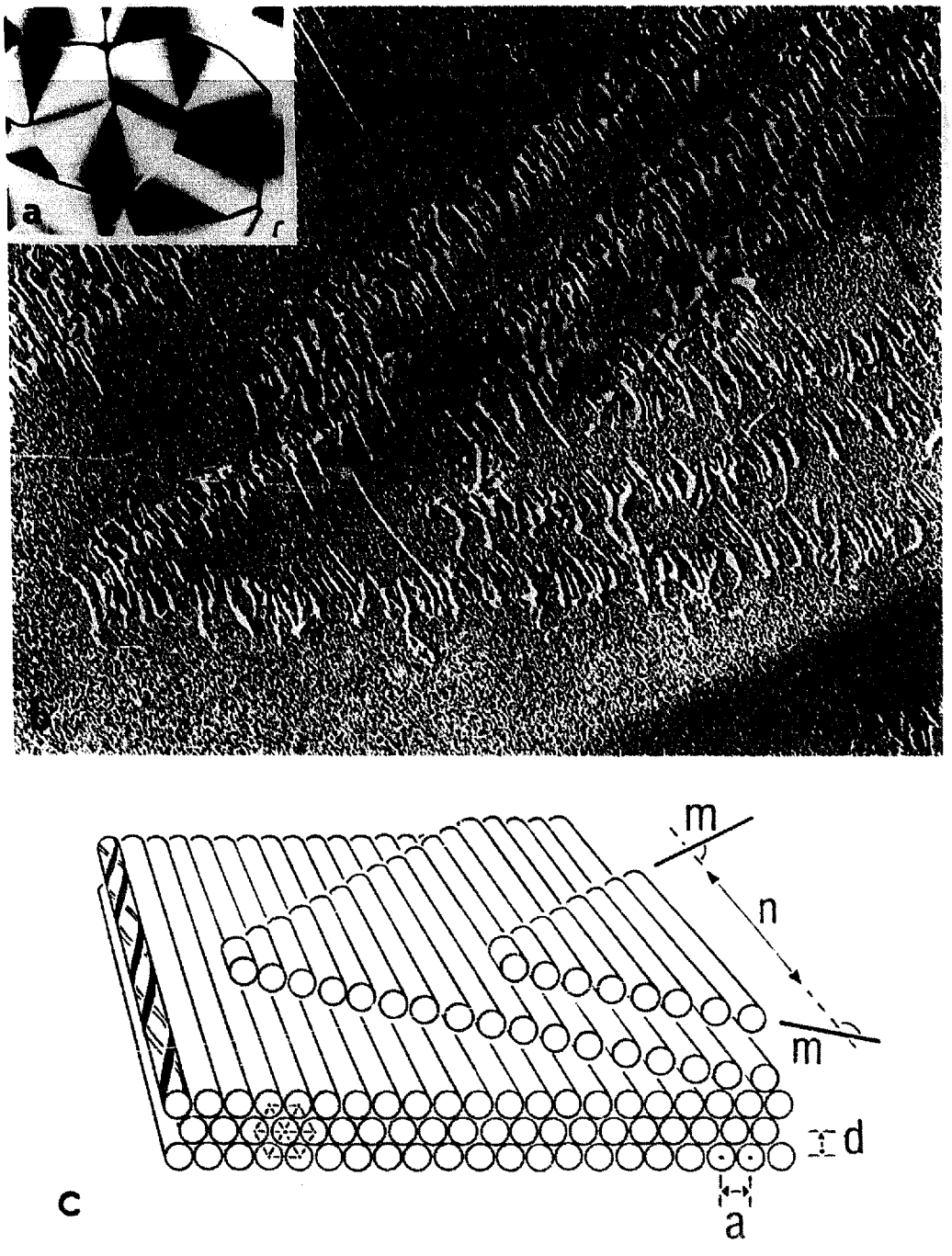


Fig. 5. Columnar hexagonal liquid crystalline phase of DNA.
(a) Textures in polarizing microscopy (crossed polars, ×190).
(b) One of the patterns observed in electron microscopy after freeze–fracture (×120 000). Superimposed layers, limited by steps (*m*) can be recognized. Each of them is made of molecules aligned in parallel along a direction (*n*).
(c) Interpretative drawing of (b). Double stranded DNA molecules, represented by cylinders, are aligned in parallel along a direction *n* and packed hexagonally, the interhelix spacing being equal to *a*. Series of layers separated by a distance *d* are revealed by the fracturing process.

and they form a sort of fringe. These planes correspond to the reticular planes of the hexagonal structure as schematically represented in fig. 5c. These patterns can be seen when the fracture plane is very close to one of the three series of main planes of the hexagonal structure [59].

Supercoiled DNA molecules were also shown recently to form a liquid crystalline phase in which supercoiled DNA molecules are hexagonally packed [60].

3.5. Other forms of densely packed DNA

Condensed forms of DNA can also be obtained in vitro by other ways:

– Polymer and salt induced (PSI) condensates first reported by Jordan et al. [61] are obtained by a phase segregation in a mixture of DNA and polyethylene glycol (PEG). Microcholesteric aggregates are formed in which DNA is in the B form. They can merge to form a classical cholesteric mesophase (see ref. [62] for a review).

– Salt precipitated forms of DNA from 60% ethanol in which DNA in the B form is packed in a hexagonal unit cell ($a = 46$ Å, $c = 34.6$ Å) [63]. This hexagonal packing (which remains local here) recalls the columnar hexagonal liquid crystalline phases obtained in aqueous solution when DNA is highly concentrated.

– The hexagonal packing of DNA was first described by Giannoni et al. [64] and Lerman et al. [65], who obtained true hexagonal crystals with sonicated DNA in alcoholic solution. Freeze–fracture experiments revealed [65] that DNA molecules were normal to the flattening plane of the hexagonal platelet.

4. Comparison between in vitro and in vivo condensed forms of chromatin

4.1. DNA concentration and helical pitch in the cholesteric phase

In vitro the helical pitch varies from 0.2 to 0.4 μm as detected by reflection of circularly polarized light in the visible (with sonicated molecules 0.05–1.9 μm and also with 500 Å DNA fragments) to more than 5 μm (as seen in polarizing microscopy) with average values of 2.2 μm [66] to 3 μm (A. Leforestier, in preparation).

In vivo, the helical pitch P varies from about 500 Å to 4500 Å, as estimated by measurements of the periodic patterns. The smallest helical pitches are observed in sperm nuclei (500–700 Å, depending on the species). In bacteria, P varies from about 1000 to 2800 Å and in dinoflagellate chromosomes from 700 to 4500 Å. In these two cases, P varies with the species and also within a given

species according to physiological conditions. These data are summarized in fig. 6.

4.2. Shape and structure of cholesteric germs

Germs of DNA which appear in vitro in equilibrium with the isotropic phase are either spherical or elongated. Dinoflagellate chromosomes are also elongated rods of condensed DNA in equilibrium with a less concentrated solution of DNA: the nucleoplasm. In both cases, the germs can be elongated with the cholesteric stratification normal to the elongation axis of the structure (fig. 7a,b). More complex shapes are also observed in vitro as well as in vivo: chromosomes and germs with buds, others showing a crack. In these cases, numerous defects are present and their detailed analysis reveals their influence on the shape of the germs [49].

The use of freeze–fracture methods makes it now possible to compare more precisely the two structures with the resolution of electron microscopy. Three different kinds of patterns can be observed in chromosomes and in liquid crystalline phases depending on the orientation of the fracture plane relative to the cholesteric stratification (either parallel, oblique or normal) [58]. The typical series of nested arcs can be recognized in both cases (fig. 7e,f).

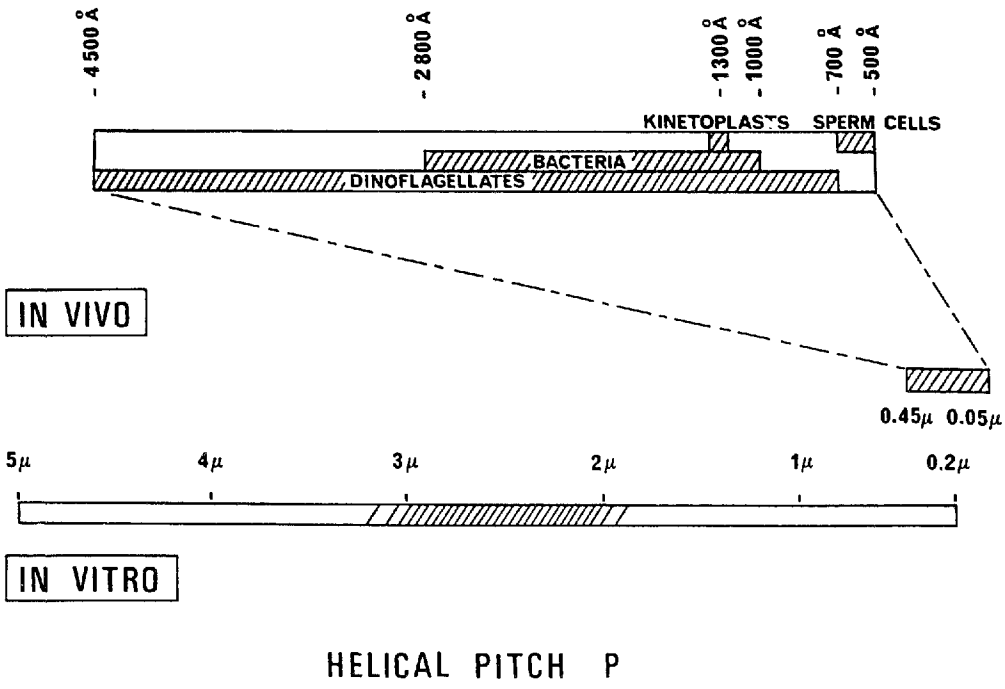


Fig. 6. Comparison of the cholesteric helical pitch P in vivo and in vitro.

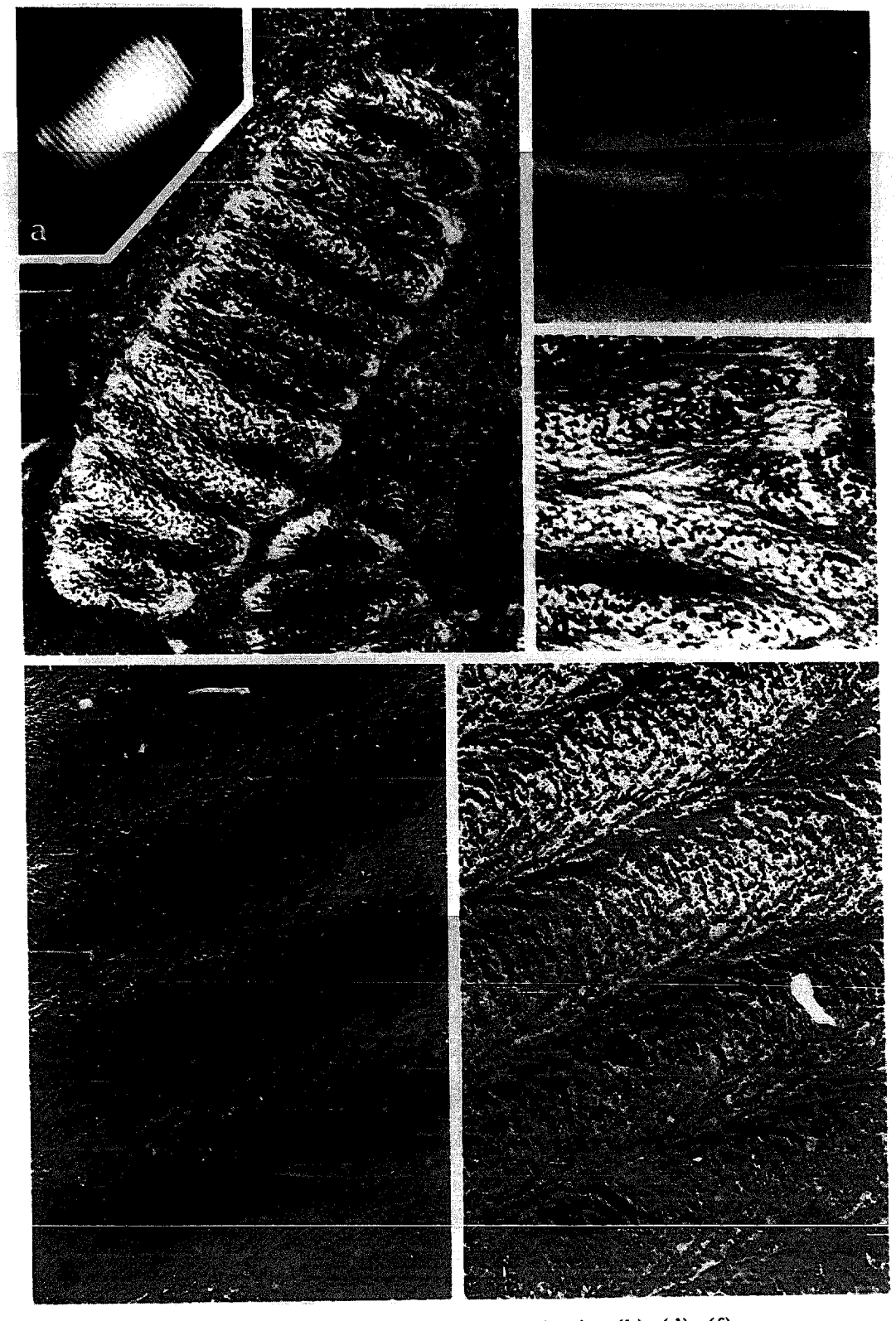


Fig. 7. Cholesteric organization in vitro (a), (c), (e) and in vivo (b), (d), (f).
(a), (b) Cholesteric germs in equilibrium with the isotropic phase. The layers are normal to the elongation axis of the structure ((a) sonicated DNA, crossed circular polars, ×900; (b) dinoflagellate chromosome, section observed in electron microscopy, ×45 000).
(c), (d) Defects $\tau^{-}\lambda^{+}$ with addition of one layer in the structure ((c) sonicated DNA, crossed circular polars, ×5000; (d) dinoflagellate chromosome, section observed in electron microscopy, ×90 000).
(e), (f) Series of nested arcs characteristic of the cholesteric organization and revealed by freeze–fracture ((e) ×7800; (f) ×90 000).

4.3. Comparative analysis of the defects

The precise analysis of the defects reveals differences between chromatin and DNA. This analysis was performed in polarizing and in electron microscopy, taking in mind that one stripe in the polarizing microscope corresponds to one layer in electron microscopy and equals $P/2$. Molecular orientations can be defined precisely in polarizing microscopy and in electron microscopy. These data are summarized in table II.

$-\pi$ rotation defects (or disclinations) are very frequent in liquid crystalline phases of DNA and they can be seen also in dinoflagellate chromosomes where they induce a strong curvature of the rod. Molecular orientations can be defined precisely in the core of the defect: in vitro, molecules lie parallel to the defect line (λ) whereas they are always normal to this line (τ) in vivo.

Uncomplete series of arcs are often observed in the dinoflagellate chromosomes (arrow in fig. 7d). They correspond to *dislocations*, which exist also in cholesteric liquid crystals and which appear in polarizing microscopy as bifurcated or interrupted illuminated lines (fig. 7c). When the defect corresponds to the addition (or removal) of one layer ($b = P/2$) the two possible situations ($\tau^-\lambda^+$) being more frequent in liquid crystals of long DNA molecules [50] and in the chromosomes (respectively 72 and 44 versus 44 and 13). These proportions seem to be reversed in liquid crystalline phases of short DNA fragments (500 Å) [74].

When the defect corresponds to the addition of several layers ($b = nP/2$), differences appear between the two systems: in chromosomes, the required number of layers is usually added successively (n defects of Burgers vector $P/2$) whereas in vitro there is one defect of a larger Burgers' vector. Neverthe-

Table II
Defects of the cholesteric structure in vitro and in vivo.

	$-\pi$ disclinations	Edge dislocations			
		$b = P/2$		$b = P$	
DNA in vitro	λ^-	$\lambda^-\tau^+$	$\tau^-\lambda^+$	$\tau^-\tau^+$	$\lambda^-\lambda^+$
		44	72	4	73
Chromosomes	τ^-	$\lambda^-\tau^+$	$\tau^-\lambda^+$	$\tau^-\lambda^+$ ($b = 2 \cdot P/2$)	$\lambda^-\lambda^+$
		13	44		2

less, we found in chromosomes two examples of $\lambda^-\lambda^+$ defects which are frequent in vitro.

Differences revealed by this analysis can probably be attributed partly to differences in the length of the DNA molecule: several meters in the chromosomes versus 0.05–1.9 μm in vitro. This hypothesis is supported by the recent analysis of the same defects in the phase made of 500 Å DNA fragments; their nature remains the same but their relative amount is different [75]. Moreover, DNA is free in vitro while it is associated with RNA, and proteins (acid and basic) in chromatin.

The biological significance of the defects is not clear. They are very numerous in some nuclei and extremely rare in others. Their presence could be related to the duplication of DNA (which occurs without decondensation of chromatin) and to the division of chromosomes.

4.4. *Left-handed helicity in vitro and in vivo*

The handedness of the cholesteric structure was demonstrated to be left-handed in dinoflagellate chromosomes and in the liquid crystalline phases either by goniometric experiments [67] or by analysis of the circular dichroism properties of the chiral structures [68, 69]. In goniometric experiments, longitudinal sections of chromosomes showing a periodic pattern devoid of arcs are tilted, which makes series of arcs to appear. The orientation of the concavity of the arcs obtained for a given tilt of the preparation can be related to the handedness of the cholesteric structure. Chromosomes of two species of dinoflagellates (*Xanthella* and *P. micans*) present a left-handed twist [2, 67] as well as the bacterial nucleoids [70]. Liquid crystalline phases of DNA, dinoflagellate nuclei and chromatin extruded from the stallion sperm nuclei were also shown to present a very strong negative circular dichroism spectrum in the absorption band of DNA indicating the presence of a long range left-handed cholesteric packing of DNA [21, 68].

4.5. *Characteristics of DNA and chromatin in highly ordered structures*

In cholesteric and hexagonal liquid crystalline phases, as well in vivo and in vitro, DNA seems to be in the B form [57]. This was shown also in PSI aggregates and in the hexagonal packing obtained in alcoholic solution [71]. The double helical structure of DNA is absolutely necessary to get this organization; denaturation of DNA, with separation of the two strands, destroys the cholesteric organization in PSI aggregates. In vivo, the B form of DNA is the classical form observed in biological materials. DNA was demonstrated to be in the B form in some cases: bacteriophages [34] and salmon sperm nuclei [72].

The linear shape of the molecule seems necessary to pack the chromatin densely. Besides, the helical shape of DNA would be responsible of the cholesteric torsion. Molecules align in parallel and a slight displacement of some of them with respect to their neighbours creates oblique grooves along which other molecules can align, which induces the formation of a twist. In this way, right-handed helices would give rise to a left-handed twist (and a left-handed cholesteric structure) and this is what we observed in vitro and in vivo. For higher concentrations, a higher packing ratio is obtained with a hexagonal packing of molecules. Then, the local twist occurring between molecules is prevented.

5. Conclusion: parallelism of DNA organization in vivo and in vitro

We have seen that a parallel can be made between the geometries of liquid crystalline DNA in vitro and the organization of chromatin in vivo.

The cholesteric organization is well represented in different types of cells. In vitro, the helical pitch varies with the DNA concentration. In chromatin, the helical pitch shows also strong variations from species to species and helical pitches ranging from 250 Å (in certain sperm nuclei) to 2000 Å (in some dinoflagellate nuclei) have been reported. Moreover, smaller variations were observed in a given cell type, which probably reflect physiological variations of the cell activity. The different values of the helical pitch correspond to different degrees of compaction of DNA, i.e. to different local concentrations of nucleic acids. These variations (of a factor 10) remain small compared to variations of the helical pitch measured in vitro (0.2–5 μm).

The decondensation of cholesteric dinoflagellate chromosomes can be induced experimentally by spreading the chromosomes onto distilled water. The cholesteric structure transforms into a double helical structure (fig. 4b, [73]). The dilution of chromatin (created by spreading) induces a transition from a cholesteric organization to a "classical" helical structure. A parallel can be made between this evolution and those observed in vitro in the reverse direction between the precholesteric stages and the cholesteric structure. It also happens sometimes that the dinoflagellate chromosomes are more or less decondensed in situ and then they often show a helical structure. We can consider that they correspond to "precholesteric" structures.

In the helical packing of DNA observed in bacteriophages [31, 74, 34], the interhelix spacing varies from about 26.4 to 27.5 Å. The interhelix spacing was shown to be in the same range in columnar hexagonal liquid crystalline phases of DNA. A hexagonal packing of DNA is also observed in trout sperm nuclei [29] and certain somatic nuclei of encysted ciliates [35, 36].

Table III

IN VITRO						
isotropic phase	⇄	precholesteric helical organization	→	cholesteric phase	⇄	columnar hexagonal phase
		——— *increase in DNA concentration* ——→				
decondensed chromatin in the interphase nucleus		decondensed chromosomes	←	dinoflagellate chromosomes		bacteriophages
		eucaryotic chromosomes with a helical shape (*Tradescantia*)		bacterial nucleoids		
				mitochondrial DNA		
				sperm nuclei (stallion)		sperm nuclei (trout, salmon)
IN VIVO						

This parallel can be summarized in the form of a table (see table III). This parallel strongly suggests that the liquid crystalline properties of DNA are involved in the process of condensation of chromatin in vivo.

Acknowledgements

We thank A. Leforestier, G. Brugerolle, J.P. Mignot and J.P. Gourret, who let us reproduce some of their micrographs.

This work was supported by a grant from ARC (Association pour la Recherche sur le Cancer).

References

[1] E. Kellenberger, E. Carlemalm, J. Sechaud, A. Ryter and G. de Haller, in: Bacterial chromatin, C. Gualerzi and C.L. Pon, eds. (Springer, Berlin, 1986) p. 11.
[2] Y. Bouligand, M.O. Soyer and S. Puiseux-Dao, Chromosoma 24 (1968) 251.
[3] P. Giesbrecht, Zbl. Bakt. I. Abt. Orig. 183 (1961) 1.
[4] A. Ryter and F. Jacob, Ann. Inst. Pasteur 107 (1964) 384.
[5] D.R. Zusman, A. Carbonell and J.Y. Haga, J. Bacteriol. 115 (1973) 1167.
[6] S. Gründ, Zbl. Met. Vet. B 21 (1974) 73.
[7] M.S. Nishimura and J. Nishiitsutsuji-Uwo, Tissue & Cell 12 (1980) 233.

[8] J.P. Gourret, Biol. Cell. 32 (1978) 299.
[9] G. Cox, Biol. Cell. 31 (1978) 117.
[10] C. Morgan, H.S. Rosenkranz, H.S. Caro and H.M. Rose, J. Bacteriol. 93 (1967) 1987.
[11] P. Dworsky and M. Schaechter, J. Bacteriol. 116 (1973) 1346.
[12] D.R. Zusman, A. Carbonell and J.Y. Haga, J. Bacteriol. 115 (1973) 1167.
[13] L. Daneo-Moore and M.L. Higgins, J. Bacteriol. 109 (1972) 1210.
[14] W. van Iterson and J.A. Aten, Antonie van Leeuwenhoek J. Microbiol. Serol. 42 (1976) 365.
[15] G. Brugerolle and J.P. Mignot, Biol. Cell 35 (1979) 111.
[16] J.K. Koehler, J. Ultrastruct. Res. 16 (1966) 359.
[17] J.K. Koehler, J. Ultrastruct. Res. 33 (1970) 598.
[18] J.E. Flechon, J. Microsc. (Paris) 19 (1974) 59.
[19] J.K. Koehler, U. Wurschmidt and M.P. Larsen, Gamete Res. 8 (1983) 357.
[20] F. Livolant, Tissue & Cell 16 (1984) 535.
[21] M.L. Sipski and T.E. Wagner, Biopolymers 16 (1977) 573.
[22] B. Picheral, J. Microsc. (Paris) 12 (1971) 107.
[23] D.M. Phillips, J. Ultrastruct. Res. 54 (1976) 397.
[24] F. Gaill and Y. Bouligand, in: Biology of Invertebrate and Lower Vertebrate Collagens, NATO ASI series, series A, Life Sciences, A. Bairati and R. Garrone, eds. (Plenum, New York, London) vol. 93, p. 267.
[25] M. Gusse and P. Chevaillier, Cytobiology 16 (1978) 421.
[26] V. Mizuhira, Y. Futaesaku, M. Ono, M. Ueno, J. Yokofujita and T. Oka, J. Ultrastruct. Res. 96 (1986) 41.
[27] M.F. Maestre, C. Bustamante, T.L. Hayes, J.A. Subirana and I. Tinoco Jr, Nature 298 (1982) 773.
[28] M. Feughelman, R. Langridge, W.E. Seeds, A.R. Stokes, H.R. Wilson, H.C.W. Hooper, M.H.F. Wilkins, R.K. Barclay and R.D. Hamilton, Nature 175 (1955) 834.
[29] V. Luzatti and A. Nicolaieff, J. Mol. Biol. 1 (1959) 127.
[30] A.C.T. North and A. Rich, Nature 191 (1961) 1242.
[31] R.M. Stroud, P. Serwer and M.J. Ross, Biophys. J. 36 (1981) 743.
[32] W.C. Earnshaw, J. King, S.C. Harrison and F.A. Eiserling, Cell 14 (1978) 559.
[33] W.C. Earnshaw and R. Casjens, Cell 21 (1980) 319.
[34] J. Lepault, J. Dubochet, W. Baschong and E. Kellenberger, EMBO J. 6 (1987) 1507.
[35] G.I. Sergeyeva, N.N. Bobyleva and R.Kh. Ibragimov, Tsitologiya (Leningrad) 29 (1987) 5 (in Russian).
[36] G.I. Sergeyeva and N.N. Bobyleva, Tsitologiya (Leningrad) 30 (1988) 1291 (in Russian).
[37] E. Delain and G. Riou, Compt. Rend. Acad. Sci. 268 (1969) 1225.
[38] E. Delain and G. Riou, Compt. Rend. Acad. Sci. 268 (1969) 1327.
[39] E.H.R. Ford, in: Human Chromosomes (Academic Press, New York, London, 1973) p. 132.
[40] L.R. Sweney, F.-H. Lam and B.L. Shapiro, J. Microscopy 115 (1979) 151.
[41] E. Polli, Biochim, Biophys. Acta 10 (1953) 215.
[42] G. Yasuzumi, G. Miyao, Y. Yamamoto and J. Yokoyama, Experientia 8 (1952) 218.
[43] C.D. Darlington and L.F. La Cour, in: The Handling of Chromosomes (Allen and Unwin, Woking, London, 1960) plate XVI.
[44] M. Ronne, Hereditas 86 (1977) 245.
[45] R. Balhorn, J. Cell Biol. 93 (1982) 298.
[46] C. Babillot, J. Microsc. (Paris) 9 (1970) 485.
[47] C. Robinson, Tetrahedron 13 (1961) 219.
[48] L.S. Lerman, Cold Spring Harb. Quant. Biol. 38 (1973) 59.
[49] F. Livolant, Eur. J. Cell Biol. 33 (1984) 300.
[50] F. Livolant, J. Phys. (Paris) 47 (1986) 1813.
[51] F. Livolant, J. Phys. (Paris) 48 (1987) 1051.
[52] Y. Bouligand and F. Livolant, J. Phys. (Paris) 45 (1984) 1899.
[53] F. Livolant and Y. Bouligand, J. Phys. (Paris) 47 (1986) 1813.

[54] R.L. Rill, Proc. Nat. Acad. Sci. USA 83 (1986) 342.
[55] T.E. Strzelecka, M.W. Davidson and R.L. Rill, Nature 331 (1988) 457.
[56] T.E. Strzelecka and R.L. Rill, J. Am. Chem. Soc. 109 (1987) 4513.
[57] F. Livolant, A.M. Levelut, J. Doucet and J.P. Benoit, Nature 339 (1989) 724.
[58] R.L. Rill, F. Livolant, H.C. Aldrich and M.W. Davidson, Chromosoma 98 (1989) 280.
[59] F. Livolant, J. Mol. Biol. 217 (1991) 165.
[60] J. Torbet and E. Dicapua, EMBO J. 8 (1989) 4351.
[61] C.F. Jordan, L.S. Lerman and J.H. Venable, Nature New Biol. 236 (1972) 67.
[62] Yu.M. Yevdokimov, S.G. Skuridin and V.I. Salyanov, Liq. Cryst. 11 (1988) 1443.
[63] D.M. Gray, S.P. Admonson, D. Lang, M. Vaughan and C. Nave, Nucl. Acids Res. 6 (1979) 2089.
[64] G. Giannoni, F.J. Padden and H.D. Keith, Proc. Nat. Acad. Sci. USA 62 (1969) 964.
[65] L.S. Lerman, L.S. Wilkerson, J.H. Venable and B.M. Robinson Jr, J. Mol. Biol. 108 (1976) 271.
[66] D.H. van Winckle, M.W. Davidson, W.-X. Chen and R.L. Rill, Macromolecules 23 (1990) 4140.
[67] F. Livolant, M.-M. Giraud and Y. Bouligand, Biol. Cell. 31 (1978) 159.
[68] F. Livolant and M.F. Maestre, Biochemistry 27 (1988) 3056.
[69] G.P. Spada, P. Brigidi and G. Gottarelli, J. Chem. Soc. Chem. Commun. (1988) 953.
[70] J.P. Gourret, Bull. Soc. Bot. Fr. 128 (1981) 55.
[71] T. Maniatis, J.H. Venable and L.S. Lerman, J. Mol. Biol. 84 (1974) 37.
[72] W.L. Kubasek, Y. Wang, G.A. Thomas, T.W. Patapoff, K.-H. Schoenwaelder, J. van der Sande and W.L. Peticolas, Biochemistry 25 (1986) 7440.
[73] F. Livolant and Y. Bouligand, Chromosoma 80 (1980) 97.
[74] M. Adrian, J. Dubochet, J. Lepault and A.W. McDowall, Nature 308 (1984) 32.
[75] A. Leforestier and F. Livolant, in preparation.